NFC Business Card CRM – Project Proposal

1. ## Overview
   This proposal outlines the development of a secure, customizable NFC Business Card CRM system that allows company admins to manage digital contact profiles linked to NFC tags. This system will empower different departments to present dynamic, updatable digital business cards that include contact information, company branding, and downloadable brochures. Built with a focus on security, user experience, and real-time data control, this platform will eliminate the dependency on third-party NFC providers and ensure full data ownership.

2. ## Key Features

- 🔐 Admin Web Portal
    - Secure login for admin and assigned roles.
    - Department and employee management.
    - Ability to generate and manage unique NFC URLs.
- 💼 Employee Profile Management
    - Profile photo, phone number, email, and department assignment.
    - Upload and manage company logo and product brochures (PDF/URL).
    - Automatic vCard generation and download link.
- 📱 Customer-Facing Digital Card Page
    - Mobile-optimized web view.
    - Auto-open from NFC tap.
    - Display of employee details, company branding, and downloadable brochure.
    - Contact-saving options (vCard).

- 🛠️ Real-Time Profile Updates

    - Admin can update any employee data or brochure at any time.
    - Instant reflection of changes upon next NFC tap (URL remains unchanged).
- 🔒 Security & Access Control
    - Encrypted storage of employee data and URLs.
    - Unique, unguessable links for each NFC card.
    - Access roles to delegate profile editing and brochure management.

3. Technology Stack (MERN)

- Frontend: React.js (with TailwindCSS for styling and responsiveness)
- Backend: Node.js with Express.js
- Database: MongoDB (cloud-based, secured)
- Authentication: JWT with role-based access
- Hosting: Vercel/Netlify for frontend, Heroku or AWS for backend
- NFC Integration: UID mapping to generated URLs

4. Milestones & Timeline

Phase	Duration	Description
UI/UX Design	1 week	Wireframes, mockups, mobile-first designs
Front-End Development	1-2 weeks	Web portal and mobile-friendly card pages
Back-End Development	2-3 weeks	APIs, database setup, NFC URL
QA Testing & Final Delivery	1 week	Bug fixing, performance checks, deployment
Total Project Duration	5-7 weeks	End-to-end delivery

5. Costing Breakdown

Component	Timeline	Cost (USD)
UI/UX Design	1 Week	$300
Front-End Development	1-2 Weeks	$600
Back-End Development	2-3 Weeks	$600
QA Testing & Delivery	1 Week	$100
Total Estimated Cost	5-7 Weeks	$1,600

6. Conclusion

This proposal outlines a secure, scalable, and user-friendly NFC Business Card CRM system tailored to your company's exact requirements. By leveraging the MERN stack, we ensure rapid development, real-time updates, and full control over your employee and customer data. This platform will empower your team with a professional, modern alternative to traditional business cards, enhanced with branding and digital brochures—all accessible with a single NFC tap.